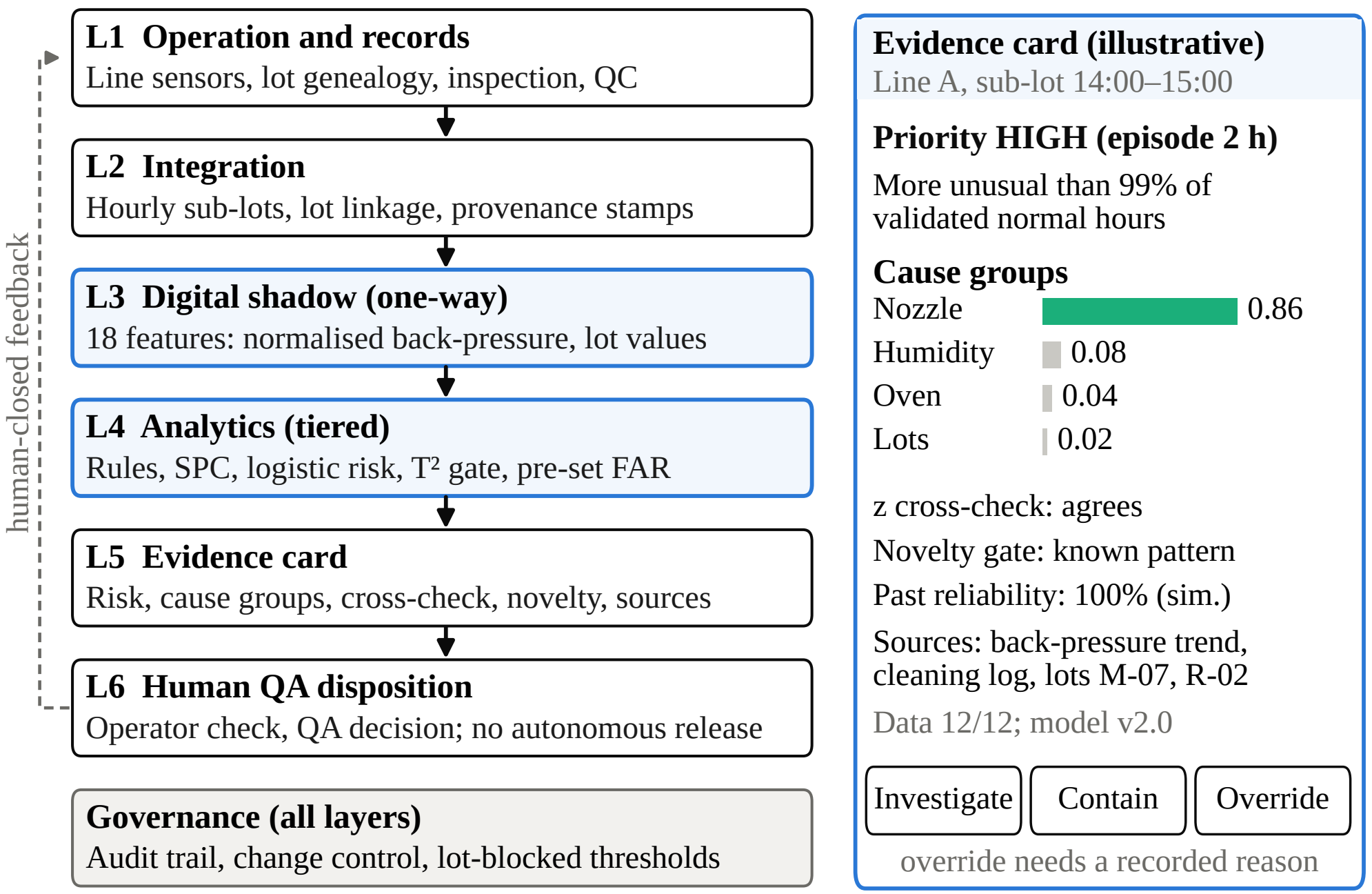
L1 Operation and records
Line sensors, lot genealogy, inspection, QC
L2 Integration
Hourly sub-lots, lot linkage, provenance stamps
L3 Digital shadow (one-way)
18 features: normalised back-pressure, lot values
L4 Analytics (tiered)
Rules, SPC, logistic risk, T² gate, pre-set FAR
L5 Evidence card
Risk, cause groups, cross-check, novelty, sources
L6 Human QA disposition
Operator check, QA decision; no autonomous release
human-closed feedback
Governance (all layers)
Audit trail, change control, lot-blocked thresholds
Evidence card (illustrative)
Line A, sub-lot 14:00–15:00
Priority HIGH (episode 2 h)
More unusual than 99% of validated normal hours
Cause groups
Nozzle 0.86
Humidity 0.08
Oven 0.04
Lots 0.02
z cross-check: agrees
Novelty gate: known pattern
Past reliability: 100% (sim.)
Sources: back-pressure trend, cleaning log, lots M-07, R-02
Data 12/12; model v2.0
Investigate
Contain
Override
override needs a recorded reason